

Aakash Luharpanchal

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Profile

Mechanical and Process Engineer with hands-on experience across mechanical design, robotics integration, automation, and precision metrology in fast-paced startup environments. Proven ability to own technical decision-making across the product life-cycle, balancing data, customer needs, cost, and execution speed. Experienced in building engineering processes, enabling teams through training and knowledge transfer, and establishing scalable technical systems that support early-stage growth and reliable product delivery

Core Competencies

- Cross-Functional Collaboration
- Automation & Robotics Integration
- Customer Project Management
- Data-Driven Decision-Making
- Product Life-cycle Management
- Precision Metrology & Measurement
- CAD & Mechanical Design
- Technical Training & Knowledge Transfer
- Systems Building & Scalable Process Architecture

Selected Highlights

- Supervised an ROS-based multi-arm robot demonstrator for in-orbit assembly, showcased at RAS 2024
- Secured Best Critical Design Review Award at the UKSEDS Olympus rover trials

Professional Experience

Zeeko Ltd, Leicester, UK — Process Engineer

Feb 2025 – Present

- Worked on ultra-precision CVD-coated ceramic components for nanometer-sensitive semiconductor manufacturing systems, applying precision metrology and data-driven process optimization to support stable and reliable system performance using Robot Processing Cell
- Led a high-precision rail engineering project, making key process and validation decisions to achieve $\leq 5\mu\text{m}$ parallelism and form tolerance targets, enabling reliable downstream system integration
- Owned day-to-day process execution while leading technical training and knowledge transfer initiatives, enabling scalable adoption of best practices across the team

ASDA, Bedford, UK — Warehouse Colleague

Apr 2024 – Feb 2025

- Inventory control and order picking with 100% accuracy
- Maintained compliance with UK workplace safety standards

Fybros Modular Pvt. Ltd., India — Senior Design Engineer

Jan 2021 – Sep 2022

- Designed and implemented a state-of-the-art relay-operated switch with integrated proximity sensor technology, achieving a robust 16-amp load capacity and recognized as a key innovation in an advanced home automation product lineup
- Guided the implementation of an economical glass plate range incorporating inventive internal metal clips that resulted in a 22% reduction in material costs and heightened assembly stability
- Produced an advanced electrical extension featuring cutting-edge sheet metal components, lowering parts count by 41% and assembly time by 80% in a pivotal company project

Cosmo Electro India Pvt. Ltd., India — Design Engineer

Jul 2018 – Nov 2020

- Conducted technical training for workers on automatic vertical tapping machines, achieving a 60% increase in production efficiency through improved process control and operator workflow optimization

- Designed and optimized modular switch components, achieving a 38% reduction in assembly complexity and aligning with lean manufacturing principles
- Coordinated all aspects of the end-to-end product development of a spike guard and the implementation of innovative cut parts that slashed total components by 56% and assembly duration by 78%
- Established a cutting-edge design management system that facilitated a remarkable 40% reduction in design errors and accelerated project delivery by 30%

Rosana Tools, India — Junior Design Engineer

Jun 2016 – Jun 2018

- Designed and deployed a 2D Cartesian robotic system to optimize sheet metal auto-feeding, improving productivity by 40%
- Programmed and operated a 3-axis CNC machine with a FANUC controller to manufacture die components and precision parts, developing custom jigs and fixtures that reduced setup time by 25%
- Devised an automatic leakage testing machine for metal tin cans used in the paint oil industry; boosted testing rate by 65%

Projects

- **Autonomous UGV (2024):** ROS2-based pallet-handling robot with radar and ultrasonic sensing
- **AI Mobile App (2024):** React Native application integrating the ChatGPT API
- **Project Mentorship (2024):** Mentored an MBA (Project Management) student on real-world project execution, translating technical constraints into actionable planning and delivery decisions
- **Autonomous Navigation (2023):** Sensor fusion and EKF-based obstacle avoidance in MATLAB
- **6-DOF Robot (2023):** Kinematic modelling and Simulink-based PID control
- **Vision Metrology (2023):** C++ and OpenCV-based system achieving 80% measurement accuracy
- **Industrial Projects (2019–2022):** Aviation tooling, 40-ton storage system, railway battery pack, and 1-ton load-bearing stand

Education

M.Sc. Robotics — Cranfield University, UK

2023

Modules: Robotics Control, AI/ML, Machine Vision, Autonomy, HRI

Group Project: Led a team of six students as the project lead, overseeing the simulation of a multi-arm robot assembly system in partnership with AIRBUS and ESA. Ingeniously built and executed an innovative magnetic plug-in for simulating the assembly process in GAZEBO and RVIZ, accomplishing it within a constrained academic timeline

Individual Project: Initiated an innovative 11-week endeavor, coordinating a study in human-robot collaboration; harnessed cutting-edge smart wearable to analyze physiological information, unveiling fresh perspectives on the dynamics of longitudinal stress and comfort

B.E. Mechanical Engineering — Mumbai University, India

2016

Modules: Mechatronics, CAD/CAM, Machine Design, FEA, Math, Kinematics, Material Technology

Group Project: Designed and fabricated an energy-generating vertical pulling exercise equipment converting human effort into electrical power

Additional Information

CAD Software: Fusion 360, SolidWorks, AutoCAD

CAM Software: SAG CAM, Zephyr CAM, Master CAM

Metrology tools and Analysis Software: Metrology Toolkit, SPSS, MATLAB, ANYSY, Form Taly-SURF, White Light Interferometer, MAHR Mar-Surf

Programming and Robotics: C++, Python, ROS, ROS2, Gazebo, ABB Robot Studio, FANUC Roboguide, Karel

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